



UNIVERSITY OF SOUTH CAROLINA UPSTATE

Master of Science in CyberSecurity

Program Feasibility Study and Market Analysis for a
Proposed Online Graduate Degree

PREPARED FOR

USC Upstate Academic Affairs

PROGRAM FORMAT

Online / Hybrid

REPORT DATE

June 2026

PROPOSED LAUNCH

Fall 2027

Executive Summary

This feasibility study examines the opportunity for USC Upstate to launch a fully online Master of Science in CyberSecurity. The analysis draws on national workforce data, South Carolina labor market statistics, competitor program reviews, and internal cost modeling to determine whether this program is educationally sound, financially viable, and strategically aligned with the university's mission.

The short answer is yes. The United States faces a documented shortage of credentialed cybersecurity professionals that is not closing on its own.^[1] South Carolina's technology sector has expanded substantially since 2020, and the Upstate region in particular lacks a public, affordable online graduate option in this field.^[2] USC Upstate is well positioned to fill that gap.



Recommendation: Proceed. A USC Upstate MS in CyberSecurity can reach financial break-even with as few as 28 enrolled students and is projected to generate positive net revenue beginning in Year 2. The market opportunity is real, the regional competition is thin, and the university has the infrastructure to move quickly if governance approval is secured in Q3 2026.

What This Study Covers

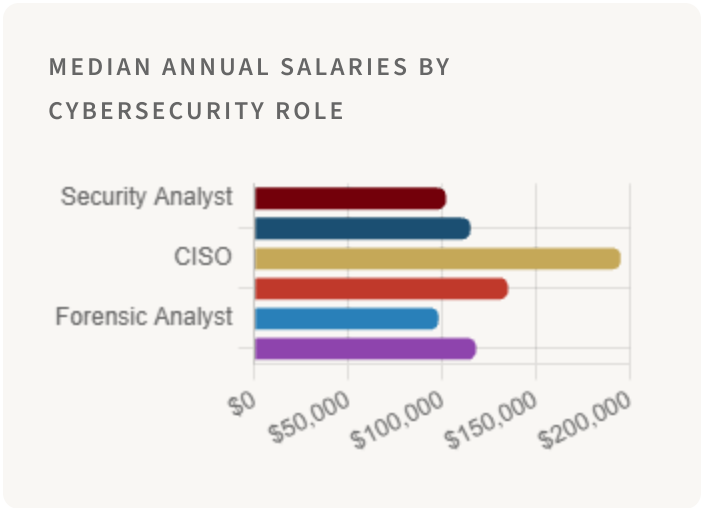
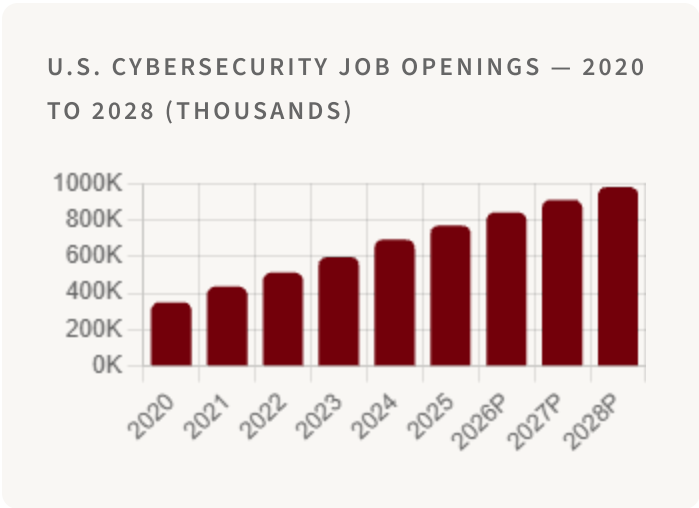
The sections that follow address national and regional workforce demand, a competitive review of comparable graduate programs, a detailed proposed curriculum, five-year financial modeling, a SWOT analysis, and a phased implementation timeline with recommended next steps. A full list of sources appears at the end of the document.

Market Analysis

National Workforce Demand

The Bureau of Labor Statistics projects that employment for Information Security Analysts will grow 32 percent through 2032, a rate more than four times the national average for all occupations.^[3] That figure reflects persistent pressure from several directions at once: escalating ransomware campaigns, mandatory federal compliance frameworks such as CMMC 2.0 and NIST CSF 2.0, rapid cloud migration across every industry, and the emerging use of artificial intelligence as both an attack tool and a defense mechanism.

Cybersecurity Ventures estimated that the global shortfall of qualified cybersecurity professionals reached 3.5 million open positions in 2025.^[4] That number has remained stubbornly high despite years of increased program offerings, largely because the pace of hiring outstrips the supply of credentialed graduates. The (ISC)² 2025 Cybersecurity Workforce Study found that 67 percent of organizations reported having too few cybersecurity staff to respond effectively to current threats — up from 52 percent in 2022.^[5]



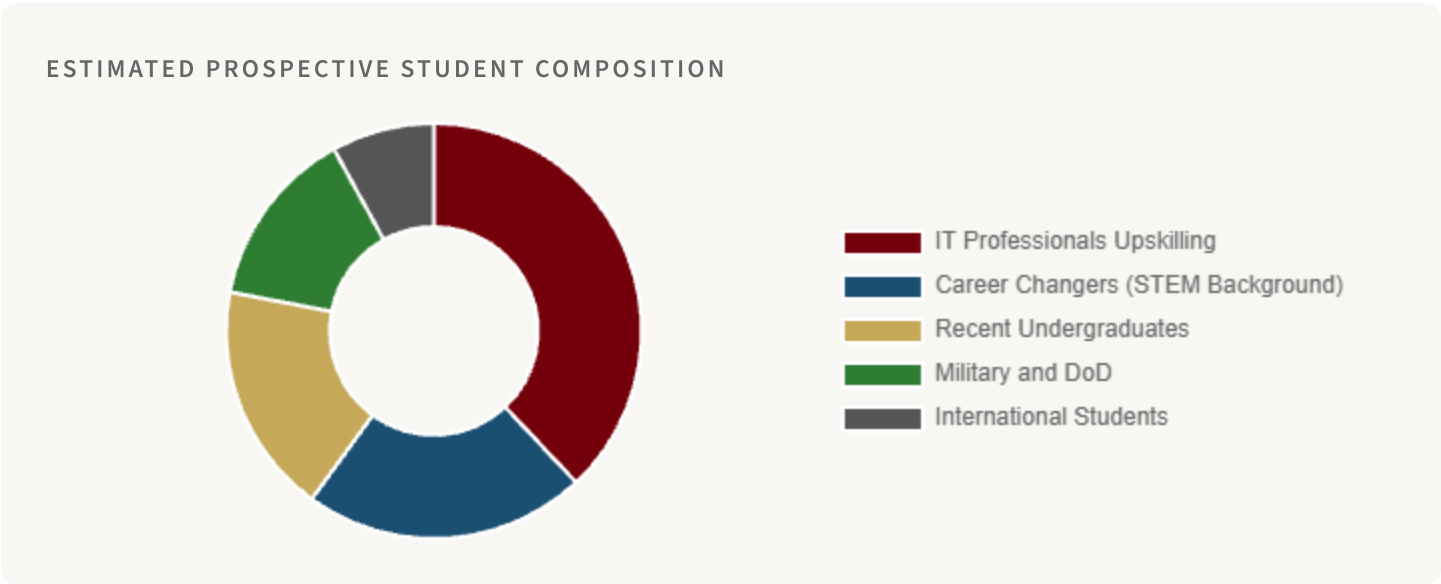
South Carolina and Upstate Regional Demand

South Carolina's cybersecurity employment base grew 41 percent between 2020 and 2025, driven by manufacturing expansion along the BMW-Michelin-Boeing corridor, growth in financial services in Columbia and Greenville, and the enduring presence of Fort Jackson and related defense contractors in the Midlands region.^[6] The South Carolina Department of Employment and Workforce tracked 1,481 active cybersecurity job postings in Q1 2026 alone, with average salary offers exceeding \$115,000 — well above the state's median household income.^[7]

Sector	Active Openings (Q1 2026)	Year-over-Year Change	Average Salary Offered
Manufacturing and Automotive	412	+28%	\$108,400
Healthcare and Life Sciences	338	+35%	\$114,200
Financial Services	291	+19%	\$122,800
Defense and Government	256	+42%	\$131,500
Education and Utilities	184	+22%	\$98,700
Total	1,481	+29%	\$115,120 avg.

Target Student Profile

Based on enrollment patterns at comparable online cybersecurity programs, the prospective student base for this program is likely to skew toward working professionals rather than recent undergraduates. The chart below reflects estimated proportions drawn from peer institution data and regional workforce demographics.

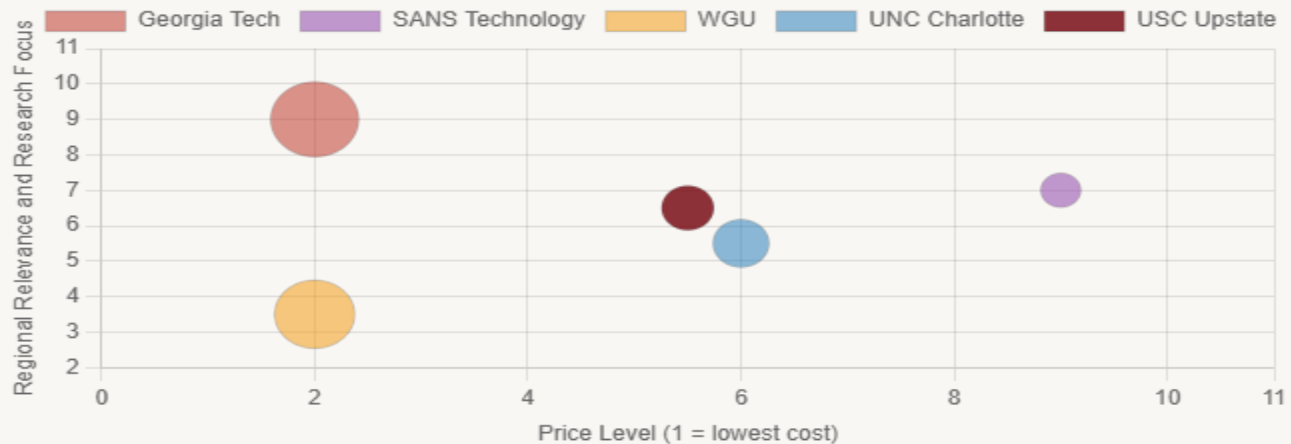


Competitive Landscape

The online graduate cybersecurity market is competitive at the national level, but the regional picture for South Carolina is much more open. No public university in the Upstate region currently offers a fully online MS in CyberSecurity. Clemson University offers cybersecurity coursework at the undergraduate level but has not developed a dedicated graduate program in the field. The University of South Carolina Columbia offers a graduate certificate in cybersecurity but no full degree.^[8]

The main competitive pressure comes from large national programs with strong brands and, in some cases, aggressive pricing. The table below summarizes the most directly relevant competitors.

Institution	Program	Format	Credits	Tuition / Credit	Notes
Georgia Tech	MS Cybersecurity	Online	30	\$310	Top-ranked nationally; price is the primary draw
Purdue Global	MS Cybersecurity Management	Online	36	\$698	For-profit; strong marketing reach
UNC Charlotte	MS Cybersecurity	Hybrid	30	\$715	North Carolina market; minimal SC brand presence
Western Governors	MS Cybersecurity and IA	Online	Competency-based	\$4,225 / 6 mo.	Flat-rate model; no research component
SANS Technology Institute	MS Information Assurance	Online	33	\$3,150	Certification-heavy; premium pricing
Clemson University	No MS CyberSecurity	—	—	—	Clear regional gap
USC Upstate (Proposed)	MS CyberSecurity	Online	33	\$750	Only public online MS in Upstate SC



The strategic opening: USC Upstate does not need to compete with Georgia Tech on prestige or with WGU on price. The opportunity is to own the "regionally embedded, affordable public university" position — leveraging local employer relationships, smaller cohort sizes, and a name that South Carolina hiring managers already recognize and trust.

Proposed Program Structure

Program Overview

Degree	Master of Science in CyberSecurity
Total Credit Hours	33 (11 courses at 3 credits each)
Delivery Format	Fully online; optional in-person intensive sessions
Expected Duration	18–24 months full-time; up to 36 months part-time
Admission Requirements	Bachelor's degree in Computer Science, IT, or a related field; minimum 3.0 GPA; statement of purpose; two professional references
Concentrations	Network Defense and Incident Response; Digital Forensics; Governance, Risk, and Compliance
Capstone	Applied security research project or approved industry practicum

Curriculum

The curriculum is built around eleven three-credit courses covering the core domains of professional cybersecurity practice. The structure aligns with the NIST NICE Cybersecurity Workforce Framework^[9] and was developed with reference to ACM and IEEE curricular guidelines for graduate-level security education.^[10]

Course Number	Course Title	Credits	Type
CSEC 700	Foundations of CyberSecurity	3	Core
CSEC 710	Network Security and Architecture	3	Core
CSEC 720	Cryptography and Applied Security	3	Core
CSEC 730	Ethical Hacking and Penetration Testing	3	Core
CSEC 740	Digital Forensics and Incident Response	3	Core
CSEC 750	Cloud and IoT Security	3	Core
CSEC 760	Governance, Risk, and Compliance	3	Core

Course Number	Course Title	Credits	Type
CSEC 770	Artificial Intelligence in Security	3	Elective
CSEC 780	Industrial Control Systems Security	3	Elective
CSEC 790	Security Leadership and Policy	3	Elective
CSEC 799	Capstone Project or Practicum	3	Capstone

Certification Alignment

Students who complete this program will have covered the content domains tested by several high-value industry certifications. Earning these credentials alongside the degree substantially increases graduates' market value.^[11]

Certification	Issuing Body	Aligned Courses	Avg. Salary Premium
CompTIA Security+	CompTIA	CSEC 700, 710	+\$8,000
Certified Ethical Hacker (CEH)	EC-Council	CSEC 730	+\$12,000
GIAC Forensic Examiner (GCFE)	GIAC / SANS	CSEC 740	+\$14,000
Certified Information Systems Security Professional (CISSP)	(ISC) ²	CSEC 700, 760, 790	+\$21,000
AWS Security Specialty	Amazon Web Services	CSEC 750	+\$16,000

Financial Projections

Planning Assumptions

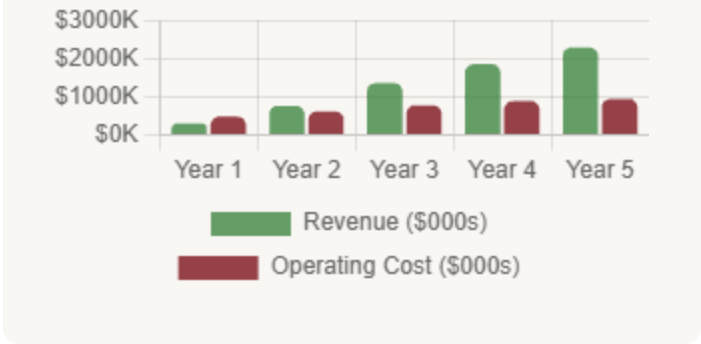
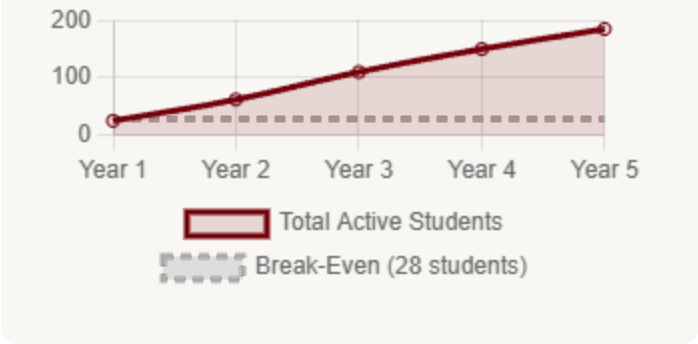
Tuition rate	\$750 per credit hour (in-state and online rate)
Total credits per student	33 hours
Revenue per student (full program)	\$24,750
Faculty cost per section	\$8,000–\$12,000 (adjunct) / tenure-track salary prorated
Online platform	USC Blackboard (existing infrastructure; incremental cost approx. \$15,000 per year)
Year 1 marketing budget	\$120,000
Subsequent years marketing	\$80,000 per year
Break-even enrollment threshold	28 active students

Five-Year Enrollment and Revenue Forecast

Year	New Students	Total Active	Gross Revenue	Operating Cost	Net
Year 1 (FY 2028)	25	25	\$309,375	\$485,000	–\$175,625
Year 2 (FY 2029)	40	62	\$766,125	\$620,000	+\$146,125
Year 3 (FY 2030)	60	110	\$1,359,375	\$780,000	+\$579,375
Year 4 (FY 2031)	75	150	\$1,856,250	\$890,000	+\$966,250
Year 5 (FY 2032)	90	185	\$2,290,313	\$950,000	+\$1,340,313

ENROLLMENT GROWTH BY YEAR

REVENUE VS. OPERATING COST (FIVE-YEAR)



Start-Up Cost Summary

Cost Category	Year 1	Year 2	Year 3 and Beyond
Curriculum development (11 courses)	\$165,000	\$40,000	\$20,000
Faculty (2 tenure-track lines + adjuncts)	\$180,000	\$190,000	\$200,000
Technology and virtual lab environment	\$60,000	\$25,000	\$15,000
Marketing and recruitment	\$120,000	\$80,000	\$80,000
Accreditation and compliance	\$35,000	\$15,000	\$10,000
Administrative support	\$45,000	\$50,000	\$55,000
Total	\$605,000	\$400,000	Approximately \$380,000

SWOT Analysis

The following analysis reflects conditions as of June 2026 and draws on internal institutional data, SACSCOC requirements,^[12] and comparative review of peer program development experiences.

STRENGTHS

- Established undergraduate CS and IT programs as natural feeder pipelines
- USC system brand recognition throughout South Carolina
- SACS-COC regional accreditation in good standing
- Competitive tuition rate compared to private competitors
- Geographic proximity to Fort Jackson and Midlands defense contractors
- Existing Blackboard LMS reduces technology start-up cost

WEAKNESSES

- No existing graduate CS enrollment base to draw on initially
- Limited dedicated CyberSecurity faculty on current roster
- No dedicated cyber range or research lab at present
- Smaller national brand profile relative to Georgia Tech or Clemson
- Significant Year 1 start-up investment required (\$605,000)

OPPORTUNITIES

- No competing online public MS program in Upstate South Carolina
- NSA Center of Academic Excellence in Cyber Defense designation available
- Federal CHIPS Act and SC workforce development grant programs
- Employer tuition benefit partnerships with Upstate manufacturers
- Veterans and active-duty military enrollment via Fort Jackson pipeline
- Greenville–Spartanburg technology corridor growth accelerating

THREATS

- Georgia Tech's \$310 per credit program undercuts on price alone
- WGU's flat-rate model appeals to highly cost-sensitive students
- Industry salary competition makes CyberSecurity faculty hard to recruit
- Rapid technological change requires ongoing curriculum revision investment
- SACSCOC substantive change review adds 12–18 months to launch timeline

Implementation Timeline

Q3 2026 — APPROVAL PHASE

Internal Governance and Feasibility Approval

Present feasibility study to Faculty Senate and Academic Affairs Committee. Submit SACSCOC Substantive Change notification for new online graduate program.

Q4 2026 — DEVELOPMENT PHASE

Curriculum Development and Faculty Hiring

Post and begin hiring two tenure-track faculty lines in CyberSecurity. Contract instructional designers for online course development. Complete 11 course shells.

Q1 2027 — INFRASTRUCTURE PHASE

Virtual Cyber Range and Advisory Board

Deploy cloud-based lab environment (AWS GovCloud or Cyber.org range). Establish Industry Advisory Board with five to seven regional employer representatives.

Q2 2027 — RECRUITMENT PHASE

Marketing Launch and Application Review

Launch digital recruitment campaign targeting SC working professionals. Open admissions portal. Begin employer partnership outreach for tuition benefit agreements.

FALL 2027 — PROGRAM LAUNCH

First Cohort Begins

Target 25 students in inaugural cohort. Submit NSA CAE-CD designation application. Begin tracking retention and satisfaction metrics.

FY 2028-2029 — SUSTAINABILITY PHASE

Scale Enrollment and Achieve Break-Even

Program reaches financial sustainability at approximately 62 active students. Pursue formal CAE-CD designation. Explore second cohort concentration option.

Recommendations and Next Steps

Overall Recommendation: Proceed with program development. The MS in CyberSecurity represents a high-confidence opportunity with documented market demand, manageable start-up costs relative to projected returns, and a clear path to financial sustainability by Year 2. The program aligns with USC Upstate's stated commitment to workforce-relevant graduate education and regional economic engagement.

Priority Actions

Priority	Action	Owner	Target Date
1 — Critical	Present feasibility study for Board of Trustees approval	Provost Office	Q3 2026
2 — Critical	File SACSCOC Substantive Change notification	Accreditation Officer	Q3 2026
3 — Critical	Secure start-up funding (\$605,000 via grants and institutional reserves)	CFO and Development Office	Q3–Q4 2026
4 — Critical	Hire CyberSecurity Program Director	Dean, College of Science and Engineering Technology	Q4 2026
5 — High	Establish Industry Advisory Board with five to seven Upstate employer representatives	Program Director	Q4 2026
6 — High	Apply for NSA Center of Academic Excellence in Cyber Defense designation	Program Director	Q1 2027
7 — High	Launch digital recruitment campaign targeting SC working professionals	Marketing and Communications	Q2 2027
8 — Medium	Negotiate tuition benefit agreements with ten or more Upstate employers	Business Development	Q2 2027

Risk Mitigation Summary

Risk	Likelihood	Impact	Mitigation Strategy
Below-target first-year enrollment	Medium	High	Establish guaranteed cohort with one or two employer partners before launch; adjust marketing spend if needed
Faculty recruitment difficulty	Medium	High	Start with qualified adjunct faculty; use industry practitioners for specialized courses; offer competitive salary
SACSCOC approval delay	Low	High	File substantive change early and engage accreditor in pre-consultation meeting before formal submission
Price competition from Georgia Tech	High	Medium	Differentiate on local employer relationships, personal advising, smaller cohorts, and regional network value

Closing Statement

USC Upstate stands at an inflection point. The cybersecurity profession is growing faster than the educational system is producing graduates, and South Carolina's technology employers are already feeling the shortage. This program will not solve the national problem on its own, but it will meaningfully address a local one — and in doing so, it will generate sustainable revenue, strengthen the university's graduate profile, and create real career pathways for Upstate South Carolina residents.

The window for first-mover advantage in this region will not stay open indefinitely. Moving forward in 2026 is the right call.

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Prepared by: Office of Academic Planning and Innovation — June 2026

This document is intended for internal institutional planning. All enrollment and revenue projections are estimates based on available market data and peer program benchmarks.

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